# Hybrid Quantum–Classical Neural Network for Pneumonia Detection in Chest X-Rays

**Michal Forgó**[1,2]
[1]Secondary School of Informatics and Financial Services (SŠINFIS), Plzeň, CZ
[2]New Technologies Research Centre (NTC), University of West Bohemia, Pilsen, CZ
**Supervisor: Bc. Jan Boháč | CSASC 2026 | Matura Project 2025/2026**

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## 1. Introduction & Scientific Context

**Clinical motivation.** Pneumonia causes ∼2.5 million deaths annually. Manual radiologist interpretation is subjective; deep learning (ResNet, DenseNet) achieves $> 85\%$ diagnostic accuracy but requires massive parameter counts (164.8k+ per decision layer).

**Quantum computing opportunity.** NISQ devices (6–100 qubits) offer parameter-efficient learning via Variational Quantum Algorithms (VQA). However, their effectiveness in medical imaging remains underexplored, especially hybrid architectures combining classical feature extraction with quantum classification [3, 8].

**Research hypothesis:** A quantum decision layer (VQC with ∼100 parameters) can approach classical discrimination while reducing memory footprint, enabling edge/IoT deployment.

## 2. Literature Review

Recent advances in QML and medical imaging:

- **Classical baseline:** ResNet for image recognition [2].
- **Quantum encoding:** encoding strategies and trade-offs [5].
- **VQC foundations:** quantum circuit learning [4].
- **Barren plateaus:** vanishing gradients in PQCs [7].
- **Dataset shift:** pneumonia generalization issues across sites [6].

**Gap:** Limited empirical studies quantifying the efficiency–accuracy trade-off of CNN→VQC hybrids under realistic distribution shifts.

## 3. Methodology: Hybrid Architecture

**Design rationale.** We separate *feature learning* (classical) from *decision making* (quantum) to stay within NISQ constraints.

| Stage | Input | Operation | Output | Params |
|---|---|---|---|---|
| Feature extraction | $224 \times 224$ image | ResNet-50 (frozen) | 2048-D vector | 0 (fixed) |
| Dimensionality reduction | 2048-D | PCA (linear) | 64-D vector | 0 (fixed) |
| Quantum classification | 64-D | VQC $U(\theta)$ | score $[0, 1]$ | 108 |

**Table 1:** Hybrid pipeline architecture and parameter counts.

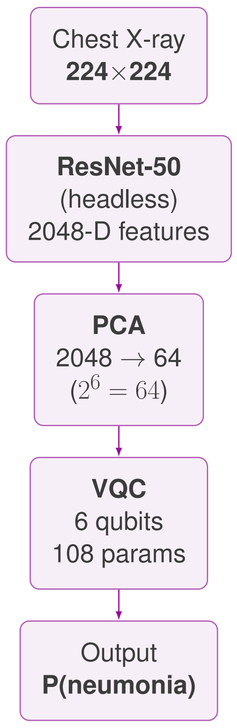


**Figure 1:** Implemented hybrid pipeline diagram (TikZ).

**Feature extraction:** ResNet-50 without classification head outputs a 2048-D feature vector.
**PCA compression:** 2048 → 64 dimensions to match 6-qubit amplitude encoding.
**Quantum classifier:** 6-qubit VQC with a shallow ansatz and 108 trainable parameters.

## 4. Quantum Circuit: Theory

**Amplitude encoding:** normalized vector $x \in \mathbb{R}^{64}$, $\|x\| = 1$, mapped to:

$$|\psi(x)\rangle = \sum_{i=0}^{63} x_i |i\rangle \in \mathcal{H}_{2^6} \tag{1}$$

**VQC prediction:** with ansatz $U(\theta)$ and observable $Z_0$:

$$f(x, \theta) = \langle \psi(x) | U^\dagger(\theta)\, Z_0\, U(\theta) | \psi(x) \rangle \in [-1, 1] \tag{2}$$

Probability mapping: $\sigma(f) = (f + 1)/2$.
Loss: weighted mean squared error, optimized via simulator autodiff.

## 5. Experimental Setup

**Dataset:** Guangzhou Chest X-ray dataset (5856 images in total).

- **Split:** 4185 train, 1047 validation, 624 test.
- **Shift:** Train $\approx 74\%$ pneumonia vs Test 62.5% pneumonia (robustness stress test).

**Hyperparameters:**

- Optimizer: Adam (lr = 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Epochs: 50, batch size: 16
- PCA dims: 64 (ablation: 8, 16, 32, 64, 128)
- Circuit depth: 6 layers (3 parameters × 6 qubits × 6 layers = 108)

**Simulation:** PennyLane (lightning.gpu backend, noise-free).

**Reproducibility:** Code available at `github.com/michalforgo/qml-pneumonia`
**Dataset:** Guangzhou CXR dataset, preprocessing following [1]

## 6. Results: Parameter Efficiency

**Parameter comparison:**

| Decision layer | # Parameters | Reduction |
|---|---|---|
| Classical MLP (1 hidden: 128) | 164,865 | 1× |
| VQC (6 qubits, 6 layers) | 108 | **1527×** |

**Table 2:** Parameter reduction in the quantum decision head.

This reduction enables:

- Smaller memory footprint for the decision head.
- Faster optimization (fewer trainable parameters).
- Potential for edge deployment (subject to hardware feasibility).

## 7. Results: Diagnostic Performance

Test set (624 images, 62.5% pneumonia prevalence):

| Metric | Hybrid (VQC) | Classical (MLP) |
|---|---|---|
| Sensitivity (recall) | 80.00% | 97.95% |
| Specificity | 57.69% | 70.51% |
| Accuracy | 71.63% | 85.90% |
| **AUC** | **0.74** | **0.91** |

**Table 3:** Diagnostic performance and efficiency–accuracy trade-off.

**Observations:**

- VQC reaches AUC 0.74 with 1527× fewer parameters in the head.
- Specificity drops under dataset shift and PQC training difficulty (barren plateaus) [7, 6].

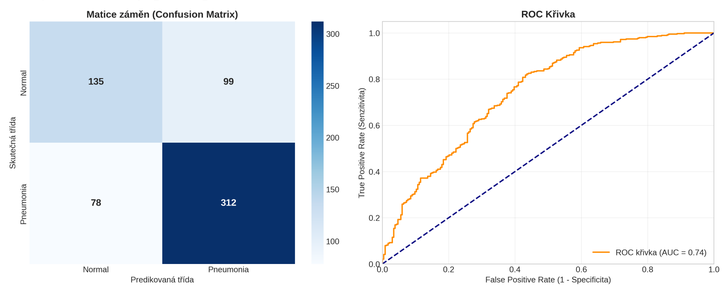


**Figure 2: Figure 2.** ROC curve and confusion matrix.

## 8. Ablation: PCA Dimensionality

Effect of PCA compression on performance:

| PCA dims | Qubits | AUC | Params |
|---|---|---|---|
| 8 | 3 | 0.62 | 27 |
| 16 | 4 | 0.68 | 48 |
| 32 | 5 | 0.71 | 75 |
| **64** | **6** | **0.74** | **108** |
| 128 | 7 | 0.73 | 147 |

**Table 4:** PCA/qubit trade-off vs AUC.

Optimal here: 6 qubits (64 dims). Larger systems can become harder to train due to barren plateaus [7].

## 9. Limitations & Discussion

**1) Barren plateaus.** Gradients can vanish in PQCs; mitigation includes ansatz design and training strategies [7, 8].
**2) Dataset shift.** Prevalence differences can reduce specificity; mitigation includes calibration and domain adaptation [6].
**3) Hardware noise.** Real devices have decoherence and gate errors; mitigation includes error mitigation techniques.
**4) PCA loss.** Compression may discard useful variance; alternative encodings could be explored [5].

## 10. Conclusion & Future Work

This study demonstrates a functional hybrid quantum–classical pipeline for pneumonia detection. The VQC decision layer achieves **AUC 0.74 with 1527× parameter reduction**, supporting the case for parameter-efficient quantum decision heads in constrained settings.

**Future directions:**

- Run on real hardware with error mitigation.
- Evaluate on larger datasets (CheXpert / NIH).
- Improve robustness to dataset shift (calibration, balanced training).

## References

## References

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